

TU CSE
TEZPUR UNIVERSITY

Spring Semester End Term Examination, 2023

CO315 (B.Tech): COMPUTER NETWORKS (CS519 for MCA backlog)

Full Marks: 60

Time: 180 mins

Figures in the right hand indicate marks for the individual question.

1	Answer any six of the followings: i. Which application/server attempts resolution of your DNS queries first? Which transport protocol is used by DNS queries? ii. What is ICMP redirect/change request? Who initiates the redirect request? iii. RARP request and response are sent unicast, multicast or broadcast? iv. How does ICMP perform clock synchronization and estimate transit time? v. How does ICMP do flow control? vi. What are the advantages of DHCP over BOOTP? vii. Differentiate between best effort forwarding and expedited forwarding for differential services? viii. How IMAP and SMTP can be distinguished? How can they co-operate while sending a mail? ix. What are the roles of SNMP proxy agent?	6x2
2	Suppose the size of congestion window of a TCP connection be 64 KB when a timeout occurs. The round trip time of the connection is 100 msec and the maximum segment size used is 4 KB. Find out the time taken (in msec) by the TCP connection to get back to 64 KB congestion window.	4
3	Consider an instance of TCP TAHOE algorithm where the window size at the start of the slow start phase is 3 MSS and the threshold at the start of the first transmission is 10 MSS. Assume that a timeout occurs during the fifth transmission. Find the congestion window size at the end of the tenth transmission.	4
4	In a telnet data transmission, a client sets up a TCP connection with a telnet server and sends a message "hey server" to the server to which server replies "hello client". Finally, the client terminates the connection. Show in a TCP flow diagram how the sequence numbers, acknowledgement numbers and TCP flags change throughout the session.	6
5	What is count-to-infinity problem? How does link state routing solve it? What are the three main additions to the header of RIPv2 as compared to RIPv1? Name a single source shortest path routing algorithm that is suitable to link state routing. How does OSPF use this algorithm with its messages? How does OSPF scale up the routing performance?	2+1+3+ 1+2+1
6	What are the characteristics of gratuitous ARP? Why is it used? Draw the DHCP client state diagram.	4+4+4
7	What are roles of Queues and control block table in UDP? What are the transport layer protocols used for real time multimedia, HTTP, DNS and email, respectively?	2+4
8	Write down the TCP messages used during a TCP connection set-up, TCP data communication and connection termination. What are the system calls in socket API that are used for connection request and connection termination request?	4+2